

Spatial Audio Final Composition

Sunshiyu Wang and Changda Ma

1 Introduction

Our inspiration comes from the characteristics of spatial audio, which has capacity to recreate and re-imagine everyday scenes through the movement of sound across shifting points and angles. In this project, we create a composition based on this idea, using spatial audio to build four environmental scenes: water, valley, city, and hall. Each shaped by its own signature sound samples and a mixing logic tailored to that setting. By mapping a MIDI controller to Reaper, the performer can switch between the four scenes in real time while continuously shaping the density, motion, directionality, and depth within each one. In this way, the work moves beyond the fixed playback of sound scenes and becomes a spatial listening system that is both performable and continuously generative. Furthermore, we explored how to port this project into Pure Data using entirely open-source tools. We have built a simple audio chain and control panel for the water scene in Pure Data, and we plan to refine the system and add more scenes and functions in the future.

2 Technology and Implementation

2.1 Overall Design Purpose

The system is structured as a performance-driven, object-based spatial audio workflow built on third-order Higher-Order Ambisonics (HOA). To create distinct sonic identities for each scene, we apply pre-processing using EQ, filtering, and chorus to the source samples, and further shape the spatialized signal using IEM’s delay and reverb plugin after the encoding stage. The system provides two output paths: a binaural decode for headphone monitoring, and a multichannel decode for the loudspeaker system in Couch 204.

2.2 Scene Overview

Four scenes are designed in total: Water, Valley, City, and Hall (see Figure 1). Each scene combines four sound objects with different spatial roles, including one continuous ambience layer, one or two

localized point sources, and one object focused on motion or depth. We also introduce pitched drone pads beneath the environmental sounds in each scene. Their harmony is tailored to the atmosphere of the scene, providing a subtle musical underlayer that supports immersion without becoming foreground melodic material. The four scenes share the same workflow and differ in their spatial character. The water scene emphasizes muffled, diffuse, and enveloping qualities; the valley scene highlights openness, depth, and discrete echoes from distant surfaces; the city scene focuses on short, dense, hard reflections and lateral motion; and the hall scene concentrates on direct sound, early reflections, and a controlled reverberant effect similar to an indoor venue. These differences are achieved by varying how the same set of IEM plugins is configured for each environment.

2.3 Spatial Audio Effect Strategy

Across the four scenes, the IEM plugins are used based on each object’s spatial role rather than as uniform effects. IEM StereoEncoder controls position, width, and motion: transient objects such as bubbles, footsteps, traffic signals, and applause remain tight and localized, while atmospheric layers are widened or slowly modulated to create envelopment. IEM FdnReverb defines each scene’s spatial character, ranging from soft and diffuse (water) to open and long (valley), short and hard-surfaced (city), and naturally reverberant (hall). IEM MultiEQ adjusts the frequency of each environment, rolling off highs underwater, preserving air in the valley, controlling low-frequency mud and high-frequency harshness in the city, and keeping musical material clear and warm in the hall. IEM Dual Delay is used only when discrete reflections are perceptually important, such as the valley walls and the hard surfaces of the city.

2.4 REAPER Routing and HOA Implementation

The spatial rendering system is implemented in REAPER using a third-order Higher-Order Ambisonics (HOA) workflow with 16 channels, in which every track and bus involved in spatial pro-

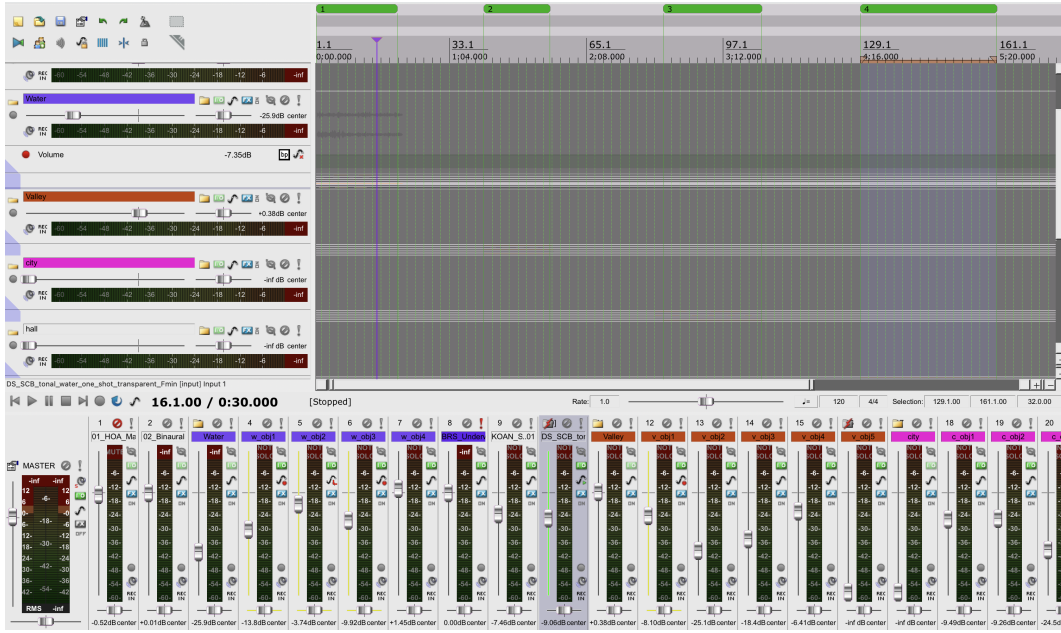


Figure 1: Reaper File Screenshot

cessing is configured with 16 channels to preserve spatial information throughout the routing chain

Each scene is organized as a folder track containing its individual sound objects as child tracks. Each object begins as mono source material and is encoded into the Ambisonic domain using an IEM encoder before being routed into its corresponding scene bus: Water, Valley, City, or Hall. Scene-level processing, such as IEM FdnReverb, IEM MultiEQ, and IEM Dual Delay, is inserted into the scene bus after the objects are combined.

After processing the four scene buses are processed, they are sent to a shared Ambisonic mix bus named “AMIX”. The AMIX track functions as the main HOA summing bus: it receives the output from the Water, Valley, City, and Hall scene busses and combines them into one complete third-order Ambisonic mix.

AMIX is then routed to two parallel decoding outputs. The first is a binaural decoding track, which uses an Ambisonic decoder to convert the third-order spatial mix into a binaural headphone output. The second is an IEM AllRADecoder, which decodes the HOA mix for playback through the spatial audio system in Couch 204.

3 Interactive design

In our REAPER-based version of the project, we use an Akai MK2 MIDI controller to control scene playback. Four midi pads are assigned to four different scenes; when a pad is triggered, the project jumps to the corresponding scene and starts play-

ing. We divide the sounds into four types: background pads, sound effects, short musical samples, and long musical drone pads. Each sound type is controlled by the same pair of MIDI knobs across different scenes. For example, the same two knobs control the spatial position of the background pad or sound effects when moving from the water scene to the valley scene. This makes the performance setup more intuitive and easier to manage. We also add a filter to the folder tracks and apply chorus to the pad sounds. Four knobs are assigned to these effects: two control the EQ’s Q and gain, and two control the chorus depth and mix level. Finally, four sliders are used to control the volume of the different scenes.

The Pure Data version is designed to provide an open-source version of the project that can be used directly without relying on our specific equipment setup (see Figure 2). However, it is much harder to build than the REAPER version, because we have to create the routing and sample playback system from scratch ourselves. In this version, we have only implemented the Water scene. We set up volume, azimuth, and elevation controls for the water pad, along with triggers for multiple whale sound samples. The spatial reverb mix is controlled by a slider. This patch is submitted alongside the project files as a starting point for our ongoing exploration of transferring the Reaper work into Pure Data.

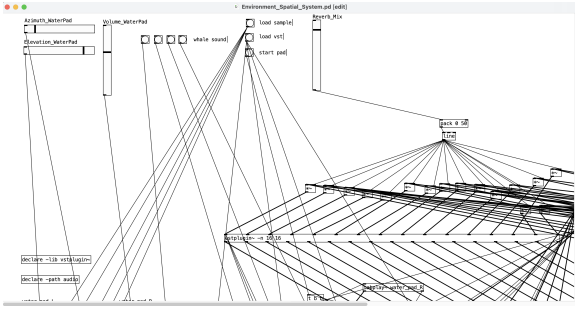


Figure 2: Pure Data version system.

4 Reflection and Sustainability

We developed two versions of the system: a Reaper version and a Pure Data version. The Reaper version was built first as our composition final project, where we successfully assigned MIDI knobs to various functions and achieved the core results we set out to create. To address sustainability and support future development, we then built a Pure Data version on top of that foundation. This version recreates the scenes and establishes an open, extensible system that can be continuously refined. Future work may include adding more scenes, creating a reusable sampler, integrating MIDI interaction, designing musical elements, and exploring further ways of interactivity.

References